Memory Forensics Report

Volatility Case Analysis – Investigation 1 & Investigation 2

1. Executive Summary

This report analyzes two memory images acquired during security incidents. Investigation 1 involves suspected banking trojan activity delivered through a malicious Adobe document. Investigation 2 involves a ransomware compromise consistent with WannaCry behavior. Volatility 3 was used to extract system details, identify suspicious processes, inspect loaded modules, analyze indicators of compromise, and attribute malware families. All screenshots referenced are stored in the evidence/screenshots directory.

2. Tooling Overview

Volatility 3 Framework was installed and verified to perform analysis on both memory samples.
Reference: Screenshot 1 – Installed Volatility

3. Investigation 1: Banking Trojan Infection

3.1 Host Machine Information

The memory image reveals:
Build Version: **2600.xpsp.080413-2111**
System Time: **2012-07-22 02:45:08**
Reference: Screenshot 2 – Case 1 Machine Info

3.2 Memory Acquisition Time

The acquisition timestamp was extracted from the memory metadata:
Acquisition Time: **2012-07-22 02:45:08**

3.3 Suspicious Process Identification

A suspicious process was detected:
Suspicious Process: **reader_sl.exe** (PID 1640)
Reference: Screenshot 3 – Case 1 Suspicious Process

3.4 Parent Process Identification

Parent Process Name: **explorer.exe**

Parent PID: **1636**

Reference: Screenshot 4 – Case 1 Process Information

3.5 Malicious Network Indicators

The memory strings show evidence of connections to banking-related domains including Chase Online, consistent with credential-harvesting activity.

Reference: Screenshot 6 – Suspicious Bank Domain Evidence

3.6 User-Agent Used by the Adversary

Extracted from memory:

User-Agent: **Mozilla/5.0 (Windows; U; MSIE 7.0; Windows NT 6.0; en-US)**

Reference: Screenshot 5 – Case 1 User-Agent

3.7 Bank Domain Confirmation

Chase Bank domain strings were found.

Answer: **Yes**

4. Investigation 2: Ransomware (WannaCry) Infection

4.1 Suspicious Process (PID 740)

A process clearly associated with WannaCry ransomware was identified.

Suspicious Process: **WanaDecryptor.exe**

Reference: Screenshot 7 – Case 2 Suspicious Process

4.2 Full Binary Path of Suspicious Process

Binary Path:

C:\Intel\ivecucqmanpnirkt615\WanaDecryptor@.exe

Reference: Screenshot 8 – Suspicious Binary Path

4.3 Parent Process Identification (PID 740)

Parent Process: **tasksche.exe**

Parent PID: **1940**

4.4 Decryptor-Related Suspicious Parent Process

Suspicious Parent PID: **1940**
This aligns with WannaCry behavior, where tasksche.exe is abused for persistence.

4.5 Malware Attribution

Based on process name, directory structure, mutexes, and DLL loads, the malware present is:
WannaCry ransomware

4.6 DLL Loaded for Socket Creation

From the module listing for the decryptor process, common networking DLLs loaded include:
ws2_32.dll (Windows Sockets DLL)
This supports malware C2 communication.

4.7 Mutex Identified

Mutexes identified include:
ShimCacheMutex, **MsWinZonesCacheCounterMutexA**, **MsWinZonesCacheCounterMutexA0**
Reference: Screenshot 9 – Mutex Indicator
These mutexes are known indicators of WannaCry variants.

4.8 Plugin to Enumerate Loaded Files

Recommended Plugin:
windows.filescan
Used to enumerate all file objects present in memory including working directory artifacts.

5. Conclusion

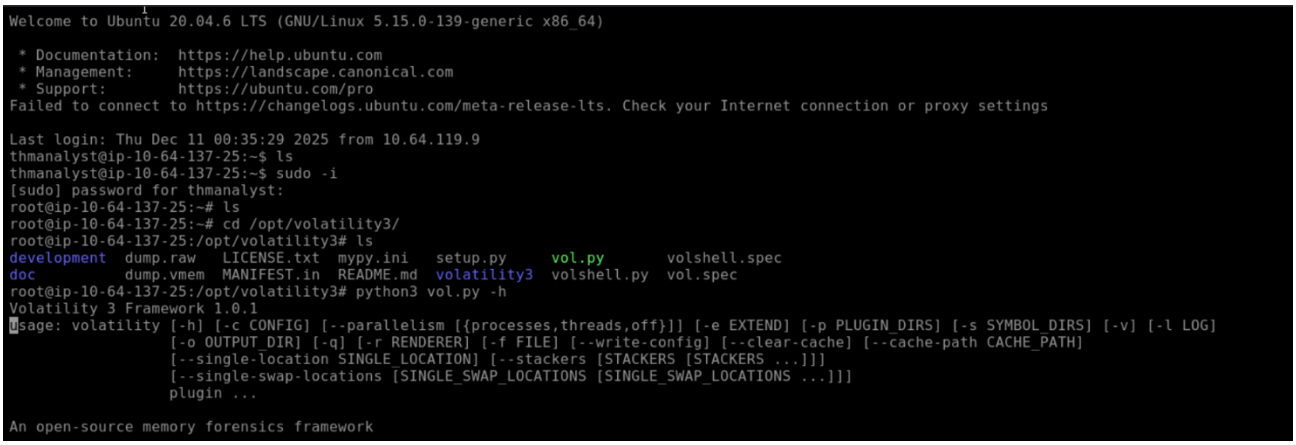
Investigation 1 confirms a banking trojan executed via Adobe Reader, using reader_sl.exe as the malicious process. The adversary connected to banking sites, likely harvesting credentials, and used a Windows-native user-agent to blend with normal traffic.

Investigation 2 confirms the presence of WannaCry ransomware, including the decryptor binary, identifying mutexes associated with this malware family, and modules required for network communication. The malicious binary and process tree match known indicators of the WannaCry campaign.

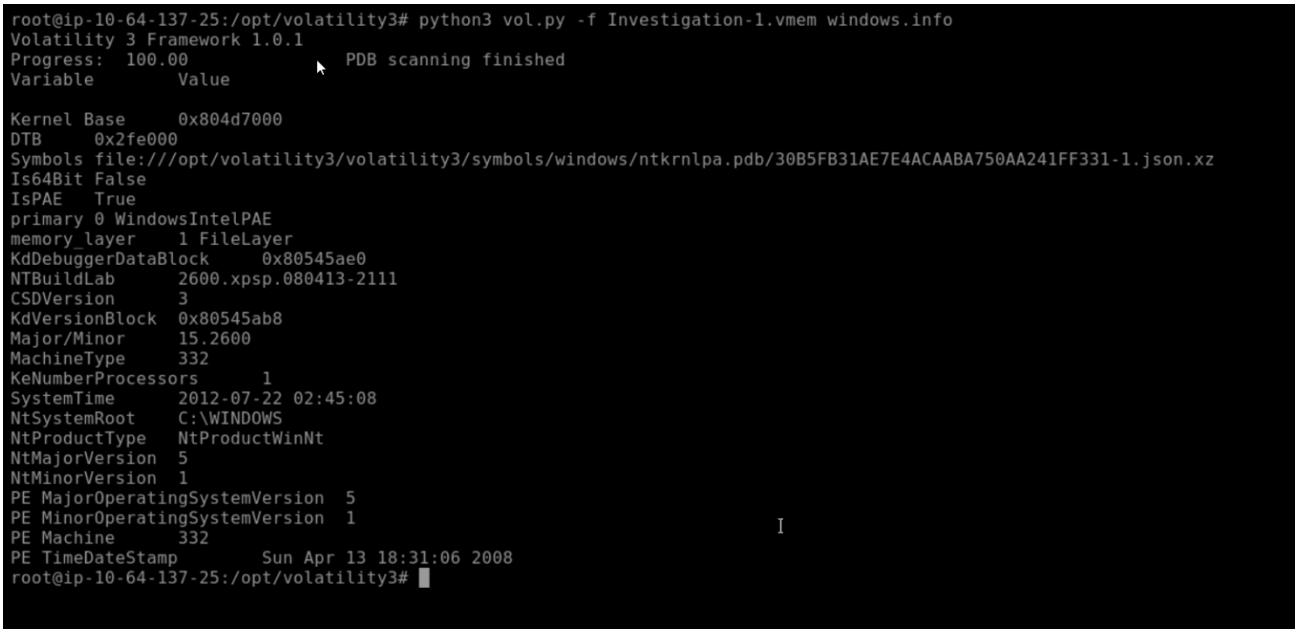
Both memory dumps demonstrate clear malicious activity and provide strong forensic evidence of compromise.

Appendix A: Evidence Screenshots

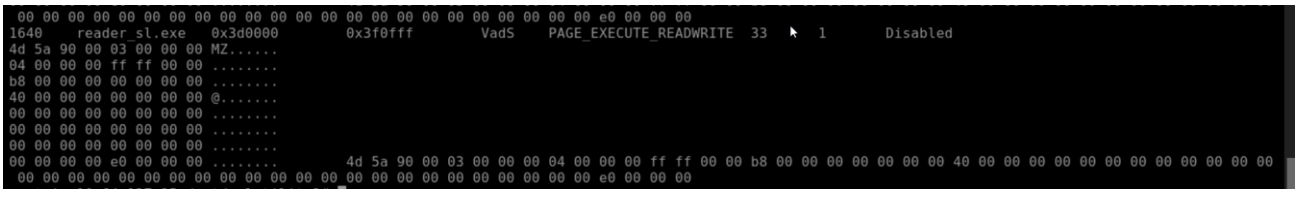
A.1 Installed Volatility



A.2 Case 1 Machine Information



A.3 Case 1 Suspicious Process



A.4 Case 1 Suspicious Process Detail

```
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime

4       0       System  0x8205bda0      53      240     N/A     False   N/A     N/A
* 368   4       smss.exe        0x8205bda0      3       19      N/A     False   2012-07-22 02:42:31.000000      N/A
** 584  368     csrss.exe       0x8205bda0      9       326     0       False   2012-07-22 02:42:32.000000      N/A
** 608  368     winlogon.exe    0x8205bda0      23      519     0       False   2012-07-22 02:42:32.000000      N/A
*** 664 608     lsass.exe       0x8205bda0      24      330     0       False   2012-07-22 02:42:32.000000      N/A
*** 652 608     services.exe    0x8205bda0      16      243     0       False   2012-07-22 02:42:32.000000      N/A
**** 1056       652     svchost.exe     0x8205bda0      5       60      0       False   2012-07-22 02:42:33.000000      N/A
**** 1220       652     svchost.exe     0x8205bda0      15      197     0       False   2012-07-22 02:42:35.000000      N/A
**** 1512       652     spoolsv.exe     0x8205bda0      14      113     0       False   2012-07-22 02:42:36.000000      N/A
**** 908        652     svchost.exe     0x8205bda0      9       226     0       False   2012-07-22 02:42:33.000000      N/A
**** 1004       652     svchost.exe     0x8205bda0      64      1118    0       False   2012-07-22 02:42:33.000000      N/A
***** 1136      1004    wuauclt.exe     0x8205bda0      8       173     0       False   2012-07-22 02:43:46.000000      N/A
***** 1588      1004    wuauclt.exe     0x8205bda0      5       132     0       False   2012-07-22 02:44:01.000000      N/A
**** 788        652     alg.exe 0x8205bda0      7       104     0       False   2012-07-22 02:43:01.000000      N/A
**** 824        652     svchost.exe     0x8205bda0      20      194     0       False   2012-07-22 02:42:33.000000      N/A
1484    1464    explorer.exe    0x8205bda0      17      415     0       False   2012-07-22 02:42:36.000000      N/A
* 1640  1484    reader_sl.exe   0x8205bda0      5       39      0       False   2012-07-22 02:42:36.000000      N/A
```

A.5 Case 1 User Agent

```
root@ip-10-64-137-25:/opt/volatility3/output# strings pid.1640.dmp | grep -i "user-agent"
User-Agent
User-Agent: Mozilla/5.0 (Windows; U; MSIE 7.0; Windows NT 6.0; en-US)
 cs(User-Agent)
USER-AGENT:
User-Agent:
root@ip-10-64-137-25:/opt/volatility3/output# strings pid.1640.dmp |grep -i "user-agent"
```

A.6 Case 1 Banking Domain Evidence

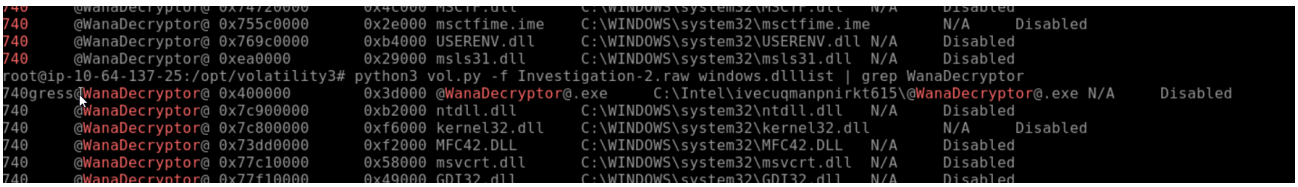
```
root@ip-10-64-137-25:/opt/volatility3/output# strings pid.1640.dmp | grep http | grep Chase
<!-- BEGIN Global Navigation table --><table cellspacing="0" cellpadding="0" border="0" class="fullwidth" summary="global navigation"><tr><td><a href=
"http://www.chase.com/" id="siteLogo"><img src="https://chaseonline.chase.com/images//ChaseNew.gif" alt="Chase Online Logo" style="margin: 17px 17px 1
7px 17px;"/></a></td><td class="globalnav"><a id="homelink" href="JavaScript:document.location.href='http://www.chase.com/';" class="globalnavlinks">C
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hase.com</a>  </td>
root@ip-10-64-137-25:/opt/volatility3/output#
```

A.7 Case 2 Suspicious Process

```
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0x823c8830      51      244     N/A     False   N/A     N/A     Disabled
348     4       smss.exe        0x82169020      3       19      N/A     False   2017-05-12 21:21:55.000000      N/A     Disabled
596     348     csrss.exe       0x82161da0      12      352     0       False   2017-05-12 21:22:00.000000      N/A     Disabled
620     348     winlogon.exe    0x8216e020      23      536     0       False   2017-05-12 21:22:01.000000      N/A     Disabled
664     620     services.exe    0x821937f0      15      265     0       False   2017-05-12 21:22:01.000000      N/A     Disabled
676     620     lsass.exe       0x82191658      23      353     0       False   2017-05-12 21:22:01.000000      N/A     Disabled
836     664     svchost.exe     0x8221a2c0      19      211     0       False   2017-05-12 21:22:02.000000      N/A     Disabled
904     664     svchost.exe     0x821b5230      9       227     0       False   2017-05-12 21:22:03.000000      N/A     Disabled
1024    664     svchost.exe     0x821af7e8      79      1366    0       False   2017-05-12 21:22:03.000000      N/A     Disabled
1084    664     svchost.exe     0x8203b7a8      6       72      0       False   2017-05-12 21:22:03.000000      N/A     Disabled
1152    664     svchost.exe     0x821bea78      10      173     0       False   2017-05-12 21:22:06.000000      N/A     Disabled
1484    664     spoolsv.exe     0x821e2da0      14      124     0       False   2017-05-12 21:22:09.000000      N/A     Disabled
1636    1608    explorer.exe    0x821d9da0      11      331     0       False   2017-05-12 21:22:10.000000      N/A     Disabled
1940    1636    taskschе.exe    0x82218da0      7       51      0       False   2017-05-12 21:22:14.000000      N/A     Disabled
1956    1636    ctfmon.exe      0x82231da0      1       86      0       False   2017-05-12 21:22:14.000000      N/A     Disabled
260     664     svchost.exe     0x81fb95d8      5       105     0       False   2017-05-12 21:22:18.000000      N/A     Disabled
740     1940    @WanaDecryptor@ 0x81fde308      2       70      0       False   2017-05-12 21:22:22.000000      N/A     Disabled
1768    1024    wuauclt.exe     0x81f747c0      7       132     0       False   2017-05-12 21:22:52.000000      N/A     Disabled
544     664     alg.exe 0x82010020      6       101     0       False   2017-05-12 21:22:55.000000      N/A     Disabled
1168    1024    wscntfy.exe     0x81fea8a0      1       37      0       False   2017-05-12 21:22:56.000000      N/A     Disabled
root@ip-10-64-137-25:/opt/volatility3# @WanaDecryptop@
```

A.8 Case 2 Suspicious Binary Path



A.9 Case 2 Mutex Indicators

